Introduction to Economics · Midterm 1

Group A6 · Universitat de Barcelona · 2026/27 · **Practice exam**

Practice exam

Name and surname

Rules

- **Practice paper:** 100 points (Part A 24, Part B 76). It has the same length, format and rules as Midterm 1 on **Thursday 19 November 2026, 10:05–11:50 (105 minutes)**; its graph questions are a little harder. Do it in one sitting, under exam conditions. It is not collected or marked; a sample solution follows on the course page.
- **Closed book.** You may use **pen, paper and any handwritten notes** (as many as you like, also photocopied or printed). **No calculators**: every number in this exam works out by hand.
- **No typed or printed text** (typed notes, slides, books). **No formula sheet** is provided. **Phones and all other electronics** are switched off and put away.
- Keep your **photo ID** on the desk; it is checked at the start.
- Any answer that is **not supported by your working** receives only partial points or none (this does not apply to the multiple-choice questions); on a graph, your drawing counts as working.
- If you use additional sheets for calculations, attach them at the end and mark clearly which question each calculation belongs to.
- Keep the pages together. Write your name on this page only.

Answering

- **Part A** (A1–A6): circle *one* letter in the answer box next to each question; exactly one option is correct; a wrong answer costs nothing. To change an answer, cross out that circle and circle the new letter:

  Answer B: (A) (B) (C) (D) Changed from B to C: (A) (B) (C) (D)

- **Part B** (B1–B6, graphs): draw on the grid of each question and label every line you draw; write the letters of an area inside it (a small area: beside it, with an arrow); write each number in its box and your calculations in the Working space. Write and draw firmly (pen, or a dark pencil): the exam is scanned.
- Points are printed next to every question: about one point per minute.

DO NOT open before start

Notation and graphs (as in class; rules and answering: page 1)

- Every grid has its own scale, printed on its axes: the price in euros on the vertical axis, the quantity on the horizontal axis. The dotted lines are halfway between the solid ones. Read and plot values on the printed scale; on some grids one square is more than one unit (look at the numbers on the axis).
- Draw straight lines through the points you are given, and label every line you draw (D, S, S + tax, . . .). Mark a point with a dot and its name (E_1). Write the letters of an area inside it; a small area: beside it, with an arrow.
- Areas are in thousand euros a week when the quantity is in thousands a week: an area of 3 (thousand units) $\times$ €2 is €6 thousand. A triangle is ½ $\times$ base $\times$ height.
- Consumer surplus (CS), producer surplus (PS), tax revenue (T), deadweight loss (DWL), social cost (SC), social value (SV); total cost (TC), fixed cost (FC), marginal cost (MC, the extra cost of one more unit), average variable cost (AVC = variable cost ÷ quantity), average total cost (ATC = TC ÷ quantity).
- Price elasticity of demand: by the midpoint method, reported as a positive number.

Extra space: write which question your working belongs to.

Part A · multiple choice, 24 points · circle one letter in the answer box

A1 [4 points]

Greta has already paid the full price, €12, for a windsurfing lesson on Saturday; the money is not refunded whatever she decides. Instead of going, she could do *one* of two jobs that day (not both): work 5 hours in a bike shop at €7 an hour, or work 2 hours as a babysitter at €13 an hour. She likes all three activities equally, and nothing else changes. What is Greta's opportunity cost of going to the windsurfing lesson?

A €35

B €26

C €61

D €47

Answer A1

(A) B C D

Scratch space (not marked)

A2 [4 points]

Which of the following shifts the demand curve for tea to the right?

A A fall in the price of tea.

B A fall in the price of the tea leaves that tea companies buy.

C A rise in the price of coffee (a substitute).

D A rise in the price of lemons (a complement).

Answer A2

A B (C) D

A3 [4 points]

A climbing wall raises the price of an entry from €6 to €8. As a result, the number of climbers per week falls from 300 to 250. Which statement is correct for this price range?

A Demand is inelastic, and total revenue falls.

B Demand is inelastic, and total revenue rises.

C Demand is elastic, and total revenue rises.

D Demand is elastic, and total revenue falls.

Answer A3

A (B) C D

Scratch space (not marked)

A4 [4 points]

Two bicycle shops on the same street, Lumo and Brillo: each sets a low or a high price, at the same time and without talking to the other. Their profits in thousand euros a week depend on both choices:

		Brillo's price	
		low	high
Lumo's price	low	Lumo 2, Brillo 2	Lumo 7, Brillo 1
	high	Lumo 1, Brillo 7	Lumo 5, Brillo 5

Which outcome is the Nash equilibrium?

A Lumo: high price; Brillo: low price.

B Lumo: high price; Brillo: high price.

C Lumo: low price; Brillo: high price.

D Lumo: low price; Brillo: low price.

Answer A4

A B C (D)

A5 [4 points]

The only massage therapist on a small island is a monopoly. The table shows its demand: the highest price at which it can sell each quantity (one price for all its customers).

Quantity (massages a day)	1	2	3	4	5
Price (euros a massage)	36	32	28	24	20

Each extra massage costs it €22 (a constant marginal cost); it has no fixed cost. Which price maximises its profit?

A €20.

B €32.

C €36.

D €24.

Answer A5

A (B) C D

Scratch space (not marked)

A6 [4 points]

In one hour, a worker in Arcadia can make 4 brooms or 4 aprons. In one hour, a worker in Boreal can make 1 broom or 3 aprons. The two countries specialise according to comparative advantage and trade brooms for aprons. At which price of one broom (paid in aprons) do both countries gain from trade?

A 5 aprons

B No price: Arcadia is better at making both goods, so trade cannot help both.

C 0.5 aprons

D 2 aprons

Answer A6

A B C D

Scratch space (not marked)

Part B · graphs, 76 points · draw on the grid and label what you draw; numbers in the boxes; calculations in the Working space

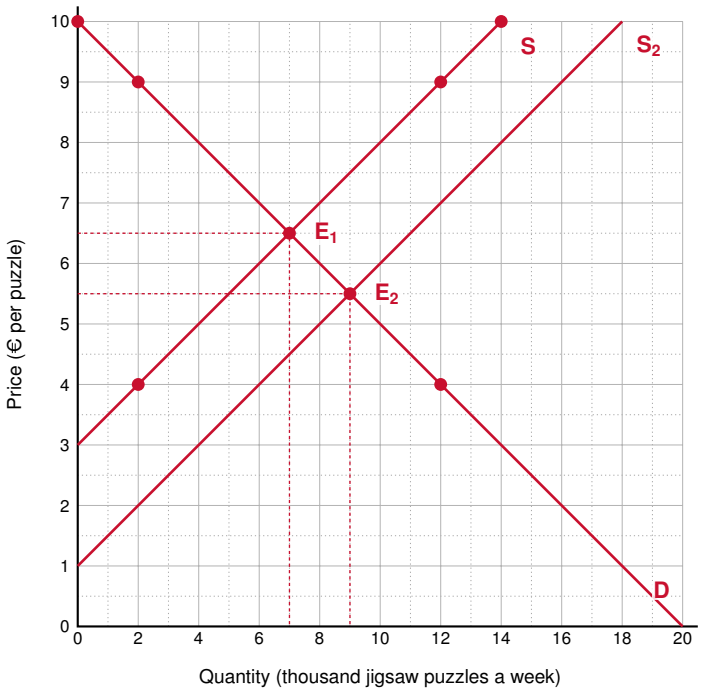
B1 · Supply and demand from a table, then a shift

[14 points; drawing 6; boxes as marked]

The market for jigsaw puzzles (quantities in thousand jigsaw puzzles a week). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.

(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.
(b) The price of cardboard, an input used to make jigsaw puzzles, falls. Making each puzzle now costs the sellers €2 less than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.
Write the price and the quantity of E_1 and of E_2 in the boxes.
(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it in the box as a positive number (a fraction or a decimal).

Price (€ per puzzle)	4	9	10
Quantity demanded (thousand jigsaw puzzles a week)	12	2	0
Quantity supplied (thousand jigsaw puzzles a week)	2	12	14



(a) Before: E_1

Price (€) [1.5]: **6.5**

Quantity (thousand) [1.5]: **7**

(b) After: E_2

Price (€) [1.5]: **5.5**

Quantity (thousand) [1.5]: **9**

(c) Price elasticity of demand, E_1 to E_2

Elasticity (a positive number) [2]: **1.5**

Working

Working

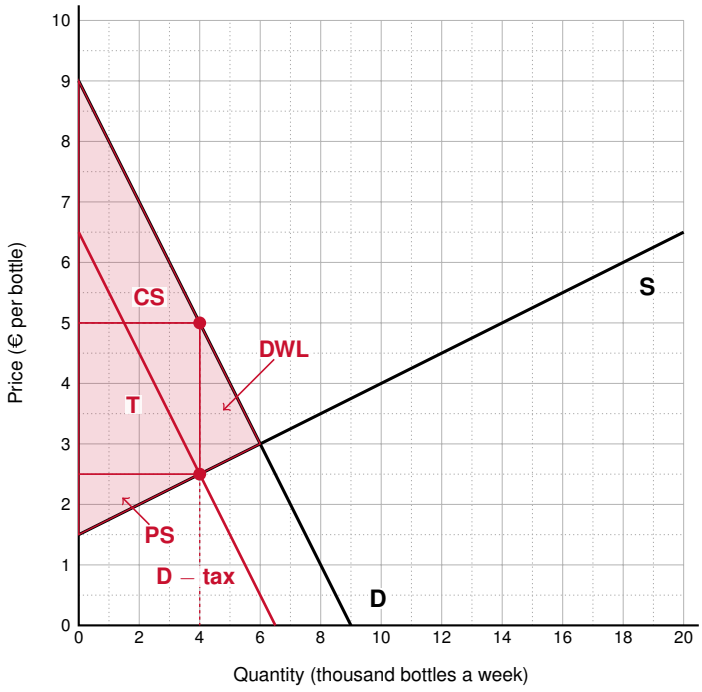
Part B · graphs, 76 points · draw on the grid and label what you draw; numbers in the boxes; calculations in the Working space

B2 · A tax: who pays what, and the areas

[14 points; drawing 4; boxes as marked]

The grid shows the market for sunscreen (sold by the bottle; quantities in thousand bottles a week): its demand curve D and supply curve S. The state puts a tax of €2.50 on every bottle bought; the buyers pay it to the state, on top of the price they pay the seller.

(a) Draw the demand curve with the tax: at every quantity €2.50 below D (what buyers will still pay the seller). Label it D − tax.
(b) Write the quantity sold, the price buyers pay in total (to the seller and the state) and the price sellers receive.
(c) Write the consumer surplus (CS), the producer surplus (PS), the tax revenue (T) and the deadweight loss (DWL) with the tax, in thousand euros a week, and write CS, PS, T and DWL inside their areas on the graph.
(d) Circle who bears more of the tax.



(b) Quantity (thousand bottles) and prices (€)

Quantity sold [1]	Buyers pay in total [1]	Sellers receive [1]
4	5	2.5

(c) Areas with the tax (thousand € a week)

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]
8	2	10	2.5

(d) Who bears more of the tax? Circle one. [1]

(Buyers) Sellers

Working

Working

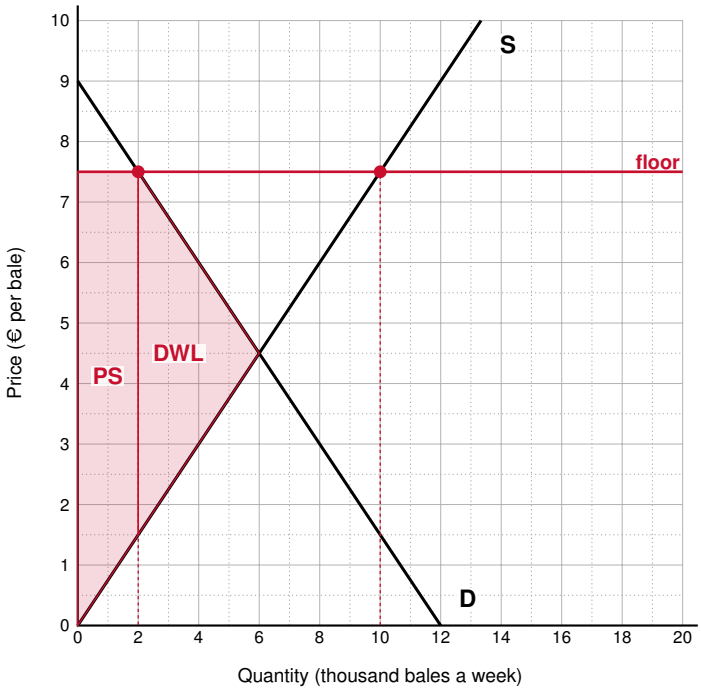
Part B · graphs, 76 points · draw on the grid and label what you draw; numbers in the boxes; calculations in the Working space

B3 · A price floor

[9 points; drawing 2; boxes as marked]

The grid shows the market for raw cotton, sold in bales (large packs; quantities in thousand bales a week): its demand curve D and supply curve S. The state sets a minimum price (a price floor) of €7.50 per bale.

(a) Draw the floor as a line and label it “floor”.
(b) Write the quantity demanded, the quantity supplied and the excess supply (the quantity sellers would like to sell but cannot) at the floor.
(c) Only the quantity buyers want is produced and sold, by the sellers with the lowest costs. Write the producer surplus (PS) and the deadweight loss (DWL) with the floor, in thousand euros a week, and write PS and DWL inside their areas.



(b) Thousand bales a week

Demanded [1]	Supplied [1]	Excess supply [0.5]
2	10	8

(c) Areas with the floor (thousand € a week)

PS [2.5]	DWL [2]
13.5	12

Working

Working

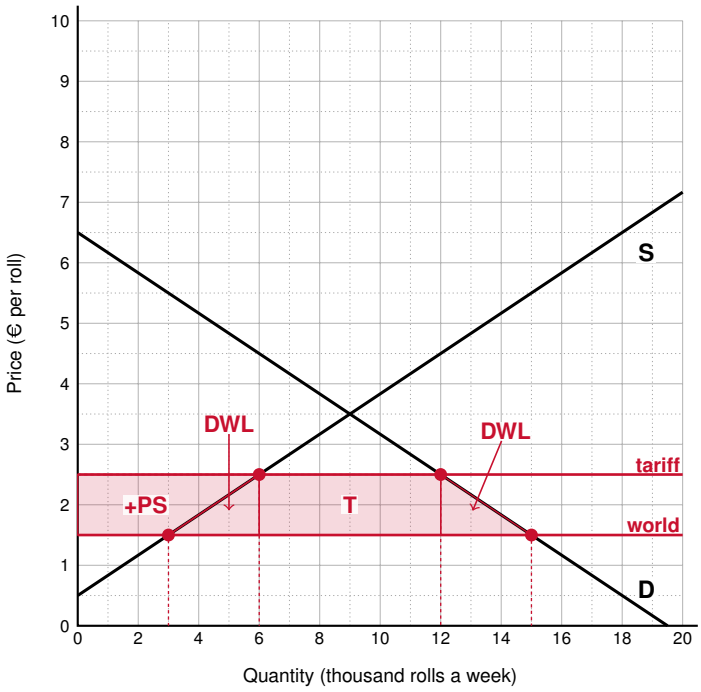
Part B · graphs, 76 points · draw on the grid and label what you draw; numbers in the boxes; calculations in the Working space

B4 · Opening to trade, then a tariff

[13 points; drawing 3; boxes as marked]

The grid shows the market for roof insulation (sold by the roll; quantities in thousand rolls a week) in a country that does not trade with other countries: its demand curve D and supply curve S. B5 uses the same market.

(a) The country opens to trade: it can buy any number of rolls at the world price of €1.50 per roll. Draw the world price as a line, label it "world", and write the country's production, consumption and imports.
(b) Then the state puts a tariff of €1 on every imported roll. Draw the price with the tariff as a line, label it "tariff", and write the price in the country, production, consumption and imports with the tariff.
(c) Write the tariff revenue (T) and the deadweight loss (DWL, both triangles together), and how much the country's producer surplus (PS) rises because of the tariff, in thousand euros a week. Write T inside its area, and DWL beside each of its two triangles, with an arrow. Shade the rise in PS on the graph and write +PS in it.



(a) Free trade (thousand rolls a week)

Production [1]	Consumption [1]	Imports [0.5]
3	15	12

(b) With the tariff (price in €, quantities in thousand)

Price [0.5]	Production [0.5]	Consumption [0.5]	Imports [0.5]
2.5	6	12	6

(c) Areas with the tariff (thousand € a week)

T [2]	DWL (total) [2]	Rise in PS [1.5]
6	3	4.5

Working

Working

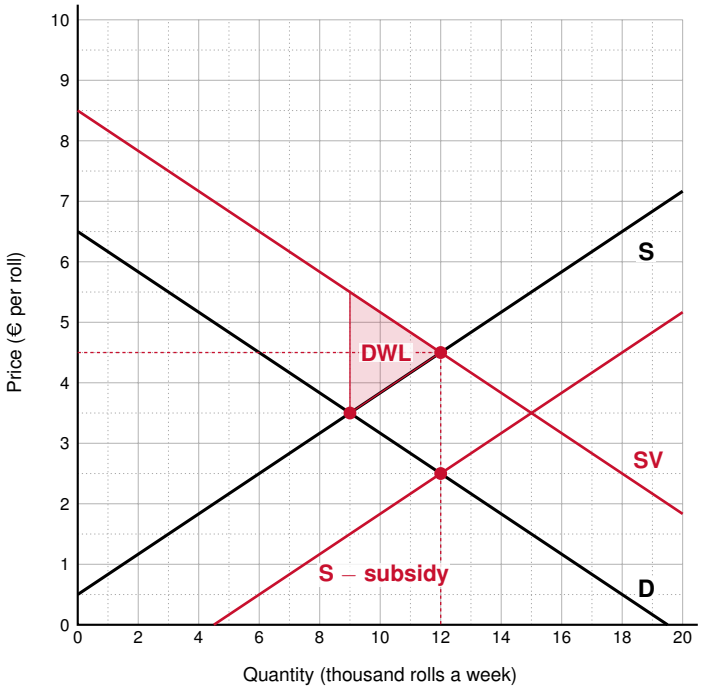
Part B · graphs, 76 points · draw on the grid and label what you draw; numbers in the boxes; calculations in the Working space

B5 · An external benefit

[10 points; drawing 4; boxes as marked]

The grid shows the market of B4 (roof insulation) without trade: D and S again. Each roll bought also benefits other people: a better-insulated home burns less heating fuel, so the air the neighbours breathe is cleaner, worth €2 per roll, a benefit the buyers do not count.

(a) Draw the social-value curve (the value of each roll to the buyer plus the benefit to others) and label it SV.
(b) Write the quantity the market sells and the efficient quantity.
(c) Write the deadweight loss (DWL) at the market quantity, in thousand euros a week, and write DWL inside its area.
(d) Write the subsidy per roll, paid to the sellers, that would bring the market to the efficient quantity. Draw the supply curve with that subsidy and label it S − subsidy (where it starts below the grid, draw the part that fits); it must cross D at the efficient quantity.



(b) Thousand rolls a week

Market quantity [1]: **9**

Efficient quantity [1.5]: **12**

(c) DWL (thousand € a week) and (d) subsidy (€ per roll)

DWL [2]: **3**

Subsidy [1.5]: **2**

Working

Working

Part B · graphs, 76 points · draw on the grid and label what you draw; numbers in the boxes; calculations in the Working space

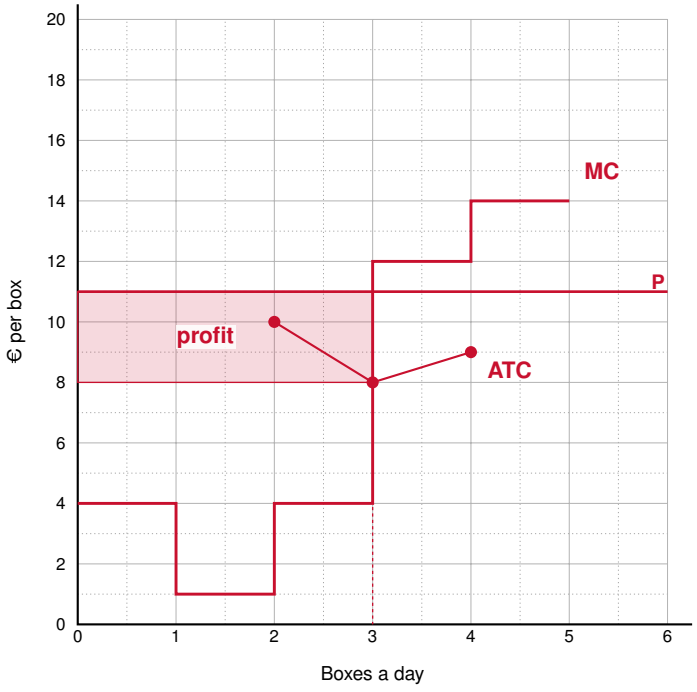
B6 · A price-taking firm

[16 points; table 5.5, drawing 4.5; boxes as marked]

A small firm sells boxes of quinces at the market price of €11 a box; it is a price taker. The table gives its total cost (TC) for 0 to 5 boxes a day.

(a) Complete the table: the marginal cost (MC) of each box (the extra cost of that box) and the average total cost (ATC) at 2, 3 and 4 boxes.
(b) On the grid, draw MC as steps: over each box, a horizontal line at that box's MC (between 2 and 3 boxes: the MC of the 3rd box). Mark the ATC at 2, 3 and 4 boxes with dots and join them. Draw the price as a line labelled P.
(c) Write the number of boxes a day that maximises the firm's profit, and that profit. Shade the profit on the graph as a rectangle and write "profit" inside it.
(d) Write the lowest price at which the firm would still produce in the short run, rather than shut down.

Boxes a day	**0**	**1**	**2**	**3**	**4**	**5**
Total cost, TC (€)	15	19	20	24	36	50
MC of that box (€)	–	**4**	**1**	**4**	**12**	**14**
ATC (€ per box)	–		**10**	**8**	**9**	



(c) Boxes a day and profit (€ a day)

Best quantity [2]: **3**

Profit [2]: **9**

(d) Price (€ a box)

Lowest price to produce [2]: **2.5**

Working

Working

Extra space: write which question your working belongs to. If you use additional sheets for calculations, attach them at the end and mark clearly which question each calculation belongs to.

Practice exam: sample solution

Introduction to Economics · Group A6

The pages before this one show every answer in red on the paper itself. Here: why. Points: Part A 24 (6×4), Part B 76.

Part A

A1 [4 points] **Answer: A**

Why. The two jobs pay 5 × €7 = €35 and 2 × €13 = €26. Only one is possible, so the best alternative given up is €35. The €12 already paid is sunk (paid whatever she decides), so it is not part of the cost of going.

A2 [4 points] **Answer: C**

Why. Coffee is a substitute for tea, so when its price rises, buyers want more tea at every price: demand shifts right. A change in the good's own price only moves along the demand curve; a cheaper input shifts supply; a more expensive complement shifts demand left.

A3 [4 points] **Answer: B**

Why. Total revenue before: €6 × 300 = €1,800; after: €8 × 250 = €2,000, so it rises. The price rose and revenue moved with the price, so the quantity changed by a smaller percentage than the price: demand is inelastic (midpoint method: change in quantity ÷ average quantity $= 50/275$, change in price ÷ average price $= 2/7$, elasticity $\frac{7}{11}$).

A4 [4 points] **Answer: D**

Why. Lumo: dominant strategy low; Brillo: dominant strategy low. In the cell Lumo: low price; Brillo: low price neither firm can raise its own profit by changing its choice alone, and this holds in no other cell. Both would earn more in the cell Lumo: high price; Brillo: high price, where both cooperate: each does its best, yet both are worse off than if both had cooperated, a prisoner's dilemma.

A5 [4 points] **Answer: B**

Why. Total revenue: 36, 64, 84, 96, 100. Marginal revenue of each extra massage: 36, 28, 20, 12, 4. Sell every massage whose marginal revenue is above the marginal cost 22: up to 2. The price is read from the demand row: €32 (revenue is largest at €20, but that ignores the cost of each massage).

A6 [4 points] **Answer: D**

Why. Opportunity cost of one broom: in Arcadia 4 ÷ 4 = 1 apron, in Boreal 3 ÷ 1 = 3 aprons. Arcadia has the comparative advantage in brooms and exports them. Both gain at any price strictly between 1 and 3 aprons, e.g. 2. (Arcadia's absolute advantage in both goods does not matter.)

Part B

B1 · Supply and demand from a table, then a shift [14 points]

(a) Plot the three points of each curve, e.g. (12; €4) on demand and (2; €4) on supply, and join them. The lines cross at E_1: price €6.50, quantity 7 thousand jigsaw puzzles a week (at €6.50, quantity demanded = quantity supplied).

(b) The price of cardboard, an input used to make jigsaw puzzles, falls: the supply curve moves down by €2 at every quantity. The new supply curve crosses the old demand curve at E_2: price €5.50, quantity 9 thousand.

(c) Only supply moved, so E_1 and E_2 lie on the same demand curve. Midpoint method: quantity, change 2 thousand, average (7 + 9)/2 = 8, so 2/8 = 0.25; price, change 1, average (6.5 + 5.5)/2 = 6, so 1/6. Elasticity = 0.25 ÷ (1/6) = 0.25 × 6 = 1.5, written as a positive number (price and quantity move in opposite directions; the minus sign is dropped): demand is elastic between these two points.

Boxes: Price (€) 6.5; Quantity (thousand) 7; Price (€) 5.5; Quantity (thousand) 9; Elasticity (a positive number) 1.5.

B2 · A tax: who pays what, and the areas [14 points]

Without the tax the market crosses at €3 and 6 thousand. The tax drives a wedge of €2.50 between the price buyers pay and the price sellers keep: the demand curve with the tax crosses S at 4 thousand bottles a week. Buyers pay €5 (read on D), sellers keep €2.50 (the crossing of D − tax with S).

CS = ½ × 4 × (9 − 5) = 8; PS = ½ × 4 × (2.50 − 1.50) = 2; T = 2.50 × 4 = 10; DWL = ½ × 2.50 × (6 − 4) = 2.5 (thousand euros a week).

Buyers pay €2 more than before, sellers keep €0.50 less: the buyers bear more of the tax (the buyers' side is less elastic: steeper here).

Boxes: Quantity sold 4; Buyers pay in total 5; Sellers receive 2.5; CS 8; PS 2; T 10; DWL 2.5; (d) Who bears more of the tax? Circle one. Buyers.

B3 · A price floor [9 points]

At the floor €7.50 buyers want 2 thousand (on D) and sellers offer 10 thousand (on S): 8 thousand stay unsold. Only 2 thousand are sold.

CS = ½ × 2 × (9 − 7.50) = 1.5. PS = the area below the floor and above S up to 2 thousand: a rectangle 2 × (7.50 − 0) minus the triangle ½ × 2 × (1.50 − 0) = 13.5. DWL = ½ × (6 − 2) × (7.50 − 1.50) = 12.

Boxes: Demanded 2; Supplied 10; Excess supply 8; PS 13.5; DWL 12.

B4 · Opening to trade, then a tariff [13 points]

At the world price €1.50 the country produces 3 thousand (on S) and consumes 15 thousand (on D): imports = 15 − 3 = 12 thousand rolls a week.

With the tariff the price in the country is €1.50 + €1 = €2.50: production 6, consumption 12, imports 6 thousand.

T = 1 × 6 = 6. DWL = ½ × 1 × (6 − 3) + ½ × 1 × (15 − 12) = 1.5 + 1.5 = 3 (thousand euros a week).

PS rises by the area between the two prices to the left of S: a trapezoid 1 × (3 + 6)/2 = 4.5.

Boxes: Production 3; Consumption 15; Imports 12; Price 2.5; Production 6; Consumption 12; Imports 6; T 6; DWL (total) 3; Rise in PS 4.5.

B5 · An external benefit [10 points]

The market sells where D meets S: 9 thousand. Counting the benefit to others, the value of each roll is €2 higher: SV is D moved up by €2. SV meets S at 12 thousand, the efficient quantity.

DWL = ½ × 2 × (12 − 9) = 3: the units between the two quantities are worth more to society than they cost, but are not bought. A subsidy of €2 per roll (the external benefit) brings the market to 12 thousand; it costs the state 2 × 12 = 24.

Boxes: Market quantity 9; Efficient quantity 12; DWL 3; Subsidy 2.

B6 · A price-taking firm [16 points]

MC = the change in TC: 4, 1, 4, 12, 14. ATC = TC ÷ boxes, at 2, 3, 4 boxes: 10, 8, 9.

A price taker sells every box whose MC is below the price €11: boxes 1 to 3 (box 4 would cost €12). Profit = 11 × 3 − 24 = €9 a day = (11 − 8) × 3, the rectangle between the price and ATC.

(d) Average variable cost (AVC) = (TC − 15) ÷ boxes: 4, 2.5, 3, 5.25, 7; the lowest is €2.50, so the firm produces in the short run at any price of at least €2.50 (below it, it shuts down and loses only its fixed cost).

Boxes: Best quantity 3; Profit 9; Lowest price to produce 2.5.